

Statistical Learning

K -Nearest Neighbor and the Bias-Variance Trade-Off

K-Nearest Neighbour

Regression Models

- Let's consider a regression model,

$$Y = f(X) + \epsilon,$$

where $E(\epsilon) = 0$ and $\text{Var}(\epsilon) = \sigma^2$.

- Collect a set of i.i.d. training data $\mathcal{D}_n = \{x_i, y_i\}_{i=1}^n$
- From $\mathcal{D}_n$, estimate the regression function as $\hat{f}$
- Predict the value of testing data Y at a target point x_0 .

k -Nearest Neighbour

- k -Nearest Neighbour (k NN) is a nonparametric method that predicts a target point x_0 with the average of nearby observations in the training data
- For regression, the prediction at a given target point x_0 is

$$\hat{y} = \frac{1}{k} \sum_{x_i \in N_k(x_0)} y_i,$$

where $N_k(x_0)$ defines a set of k samples from the training data (in terms of their feature values) that are closest to x_0 .

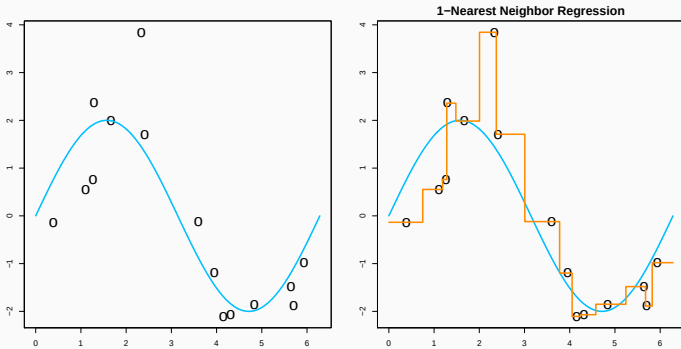
- Can also be used for classification

Example

Data with only 1 feature from uniform $[0, 2\pi]$. The true model (blue) is

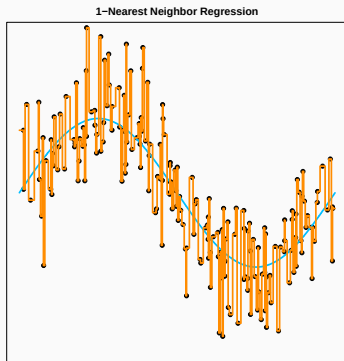
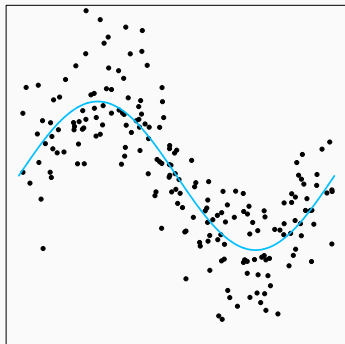
$$Y = 2 \sin(X) + \epsilon,$$

where ϵ is standard normal error. We fit the data with 1NN.



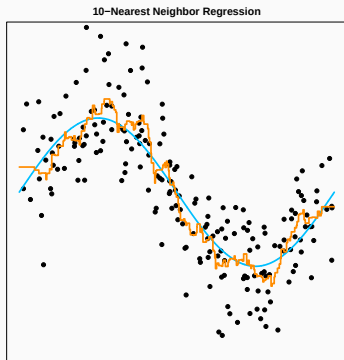
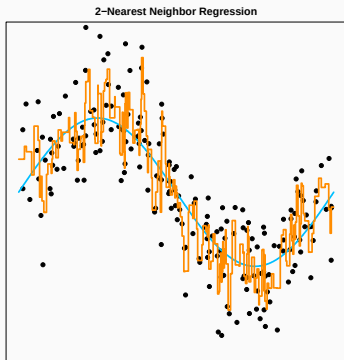
k -Nearest Neighbour in Regression

Simulate 200 observations, and see how the model changes over k .



k -Nearest Neighbour in Regression

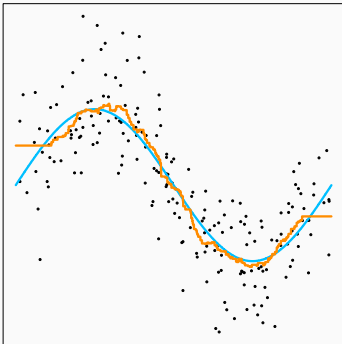
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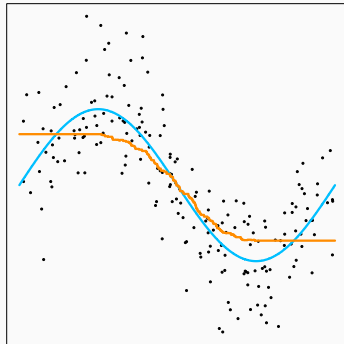
k -Nearest Neighbour in Regression

The model becomes “smoother” as k increases. However, this eventually deviates from the truth if k is too large.

33-Nearest Neighbor Regression



100-Nearest Neighbor Regression



The Bias-variance Trade-off

- We can see when k is small, the estimated model is unstable.
- Also, each time we observe a new training data, we may get a very different estimation. (due to the closest sample and ϵ)
- The statistical quantity to describe this property is the variance of the estimator $\hat{f}$.
- For a target point x_0 , variance of $\hat{f}(x_0)$ is

$$\text{Var}(\hat{f}(x_0)) = \mathbb{E}[(\hat{f}(x_0) - \mathbb{E}\hat{f}(x_0))^2]$$

- When k is large, the estimated model eventually deviates (systematically) from the truth.
- The statistical quantity to describe this property is the bias of the estimator $\hat{f}$.
- For a target point x_0 , bias of $\hat{f}(x_0)$ is

$$\text{Bias}(\hat{f}(x_0)) = f(x_0) - \mathbb{E}\hat{f}(x_0)$$

An accurate prediction

- Using squared-error loss, the prediction error is

$$\begin{aligned}\text{Err}(x_0) &= \mathbb{E}_{\mathcal{D}_n, Y_0} \left[(Y_0 - \hat{f}(x_0))^2 \right] \\ &= \mathbb{E}_{\mathcal{D}_n, Y_0} \left[(Y_0 - f(x_0) + f(x_0) - \mathbb{E}_{\mathcal{D}_n} \hat{f}(x_0) + \mathbb{E}_{\mathcal{D}_n} \hat{f}(x_0) - \hat{f}(x_0))^2 \right] \\ &= \dots \\ &= \underbrace{\mathbb{E}_{Y_0} \left[(Y_0 - f(x_0))^2 \right]}_{\text{Irreducible Error}} + \underbrace{\left(f(x_0) - \mathbb{E}_{\mathcal{D}_n} \hat{f}(x_0) \right)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}_n} \left[(\hat{f}(x_0) - \mathbb{E}_{\mathcal{D}_n} \hat{f}(x_0))^2 \right]}_{\text{Variance}}\end{aligned}$$

- All the cross terms are zero

An accurate prediction

- The prediction error is a sum of three terms
- $E[(Y - f(x))^2] = \sigma^2$ is the **irreducible error** term that cannot be avoided, because we cannot predict ϵ
- Of course, we want to minimize both **Bias²** and the **Variance**, however, this is not always possible...

An accurate prediction

- The first instinct is that one should minimize the Bias² term
- **This instinct is wrong!** — This is usually at the expense of high variance, which eventually damages the prediction performance...
- We already see this with 1NN

An accurate prediction

- 1NN has small Bias², as the closest neighbor converges to the target point x_0 as $n \rightarrow \infty$
- However, the variance is large because the estimator only uses one observation, roughly

$$\mathbb{E}[(\hat{f}(x_0) - \mathbb{E}\hat{f}(x_0))^2] = \mathbb{E}\epsilon^2 = \sigma^2.$$

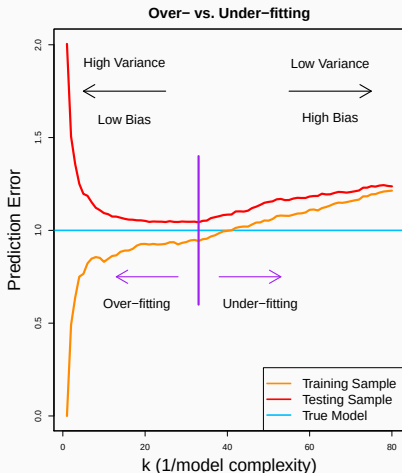
- If we use more “neighbouring” points, say k , the variance would reduce to approximately σ^2/k . But the Bias² will increase as neighbours are far away from x_0 .

Model Complexity

- A related concept is the model complexity
- In linear regression, this is simply the degrees of freedom
- For k NN, k determines the model complexity
- e.g., when $k = N$, this is just the sample mean, and the model is very simple

Model Complexity, over- and under-fitting

- Model complexity $\uparrow$ (small k) $\longrightarrow$ Bias² $\downarrow$ and Variance $\uparrow$
- Model complexity $\downarrow$ (large k) $\longrightarrow$ Bias² $\uparrow$ and Variance $\downarrow$

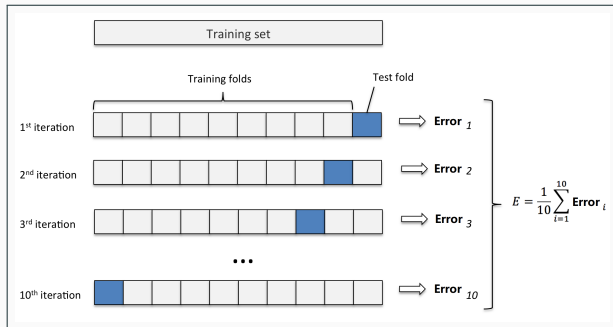


Prevent over-fitting

- As we can see, model complexity, bias-variance trade-off and over- and under-fitting are usually related concepts
- Over-fitting happens when a model performs well on the training data, but not on the testing data
- How to choose k to prevent over-fitting?

k -fold Cross-validation

- Randomly split the data into 10 portions
- Fit the model using 9 portions as training data
- Calculate the testing error using the remaining portion
- Alternate the testing set and average all testing errors



k -fold Cross-validation

- A k -fold cross-validation is random and the result can be affected by the randomness
- Cross-validation has many variations
 - We could repeatedly run k -fold cross-validation several times
 - Repeated random sub-sampling is another choice
 - Leave-one-out cross-validation is deterministic, but takes longer to run

Degrees-of-freedom and Model Complexity

- Degrees-of-freedom and model complexity are related concepts, and they can be used to prevent over-fitting
- For k NN, the tuning parameter k directly controls both
- For some other models, a penalized framework is used to control complexity

$$\arg \min_f \text{loss}(f) + \lambda \text{complexity}(f)$$

- More examples: Lasso, Ridge, tree models, etc.

Degrees-of-freedom

- Sometime the model complexity can be measured using the **degrees-of-freedom**
- In linear regression, the degrees-of-freedom is the number of variables used in the model
- A more general definition is

$$\text{df}(\hat{f}) = \frac{1}{\sigma^2} \sum_{i=1}^n \text{Cov}(\hat{Y}_i, Y_i)$$

- **Treat $X_i = x_i$'s as fixed values**, not random
- $1/\sigma^2$ takes care of the variance of the random error term

Example

- If we let $\hat{\mathbf{Y}} = (\hat{Y}_1, \dots, \hat{Y}_n)$ and $\mathbf{Y} = (Y_1, \dots, Y_n)$, we can rewrite the definition as

$$\text{df}(\hat{f}) = \frac{1}{\sigma^2} \text{Trace}(\text{Cov}(\hat{\mathbf{Y}}, \mathbf{Y}))$$

- This can be convenient for linear regression

Example

We can easily verify several cases:

- For 1NN, $\text{df} = n$
- If $\hat{y}_i = \bar{y}$, i.e., $n\text{NN}$, then $\text{df} = 1$
- For linear regression, $\text{df} = p$
- For $k\text{NN}$, $\text{df} = n/k$

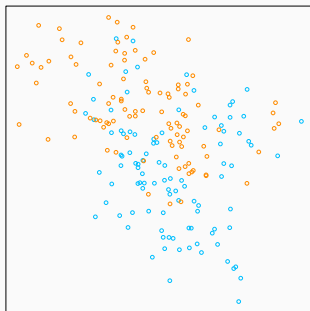
k NN for Classification

k NN for Classification

For hard classification, the most prevalent class in $N_k(x_0)$ is used

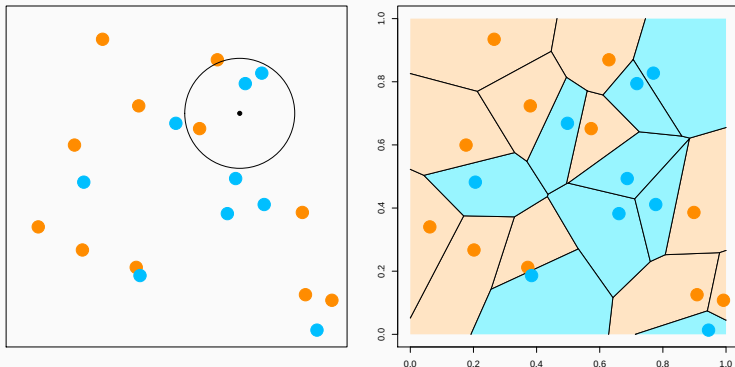
$$\hat{y} = \arg \max_{c \in C} \sum_{x_i \in N_k(x_0)} \mathbf{1}\{y_i = c\},$$

An example from the ESL textbook. (BLUE = 0, ORANGE = 1)



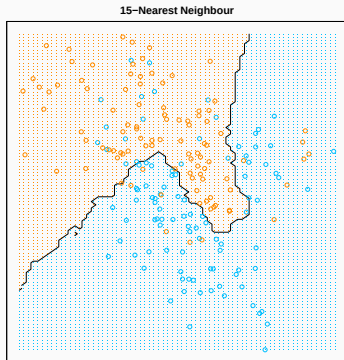
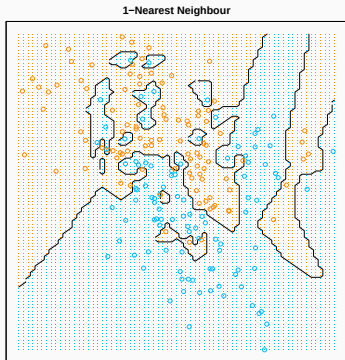
Voronoi Tessellation

Similar to the regression case, the k -NN classification model does majority vote (the most prevalent class) within the neighborhood of a target point x . 1NN plot is a Voronoi tessellation



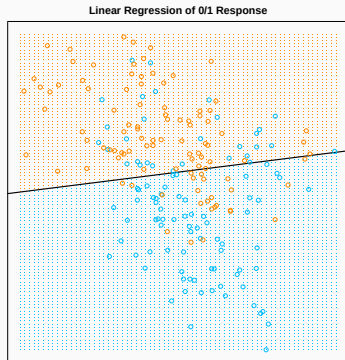
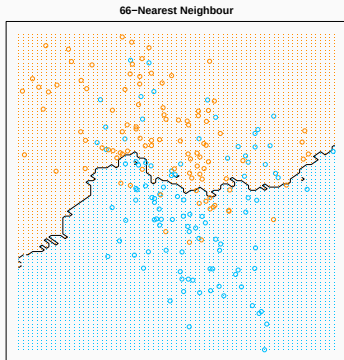
Example

We fit k -NN classification model to the example. Of course, we would not expect 1NN to perform well...



k -Nearest Neighbour in Classification

As we further increase k , the model tends to be less complex.
Compare 66NN with a linear model.



1NN Error Bound

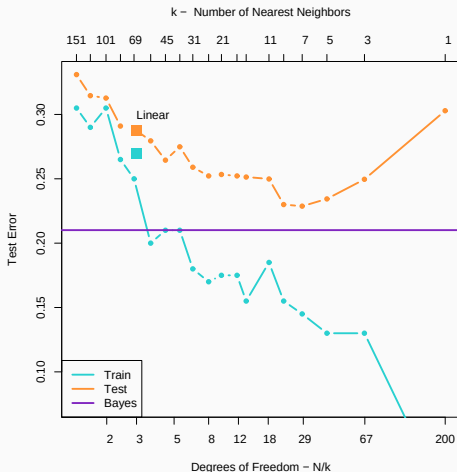
- 1NN error is no more than twice of the Bayes error, as $n \rightarrow \infty$
- As $n \rightarrow \infty$, we have $d(x_0, x_{1nn}) \rightarrow 0$, where x_{1nn} is the closest neighbor of x_0 . and we may assume that $P(Y|x_{1nn}) \rightarrow P(Y|x_0)$
- The error of 1NN is

$$\begin{aligned} & P(Y = 1|x_0)[1 - P(Y = 1|x_{1nn})] + [1 - P(Y = 1|x_0)]P(Y = 1|x_{1nn}) \\ & \leq [1 - P(Y = 1|x_{1nn})] + [1 - P(Y = 1|x_0)] \\ & \approx 2[1 - P(Y = 1|x_0)] \\ & = 2 \times \text{Bayes Error} \end{aligned}$$

- This is a very crude bound, but it shows that if the noise is small, 1NN may be reasonable.

k -Nearest Neighbour in Classification

An “U” shaped prediction error curve is again observed for the testing sample (Figure from ESL):



Remarks

- k NN vs. linear model
- Distance measure
- Computational issue
- Curse of dimensionality
- Double descent

k -Nearest Neighbour vs. Linear Regression

- The goal is to approximate $f(x) = E(Y|X = x)$
- Linear regression makes a structural assumption: f is linear.
 - **low variance**: Number of parameters is p (fixed); we know that when sample size n grows, the variance of $\hat{\beta}$ is $\propto 1/n$.
 - **high bias** (underfit): linear assumption is very restrictive
- k NN makes no assumption on f , except some smoothness.
 - **low bias** (overfit): flexible and adaptive. It can be shown that as if $k \rightarrow \infty$ and $k/n \rightarrow 0$, k NN is consistent.
 - **high variance**: number of parameters for k NN is roughly n/k ;

Distance measures

- Closeness between two points needs to be defined based on some distance measures
- By default, we use Euclidean distance (ℓ_2 norm) for continuous variables

$$d^2(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_2^2 = \sum_{i=1}^p (u_i - v_i)^2$$

Hence the neighbourhood is not invariant to the scaling of the variables.

Scaling issues

- We often scale the variables marginally when using k NN, so that the distance is

$$d^2(\mathbf{u}, \mathbf{v}) = \sum_{j=1}^p \frac{(u_j - v_j)^2}{\sigma_j^2}$$

where σ_j^2 is the variance of variable j .

- Mahalanobis distance is also scale-invariant and takes care of correlation

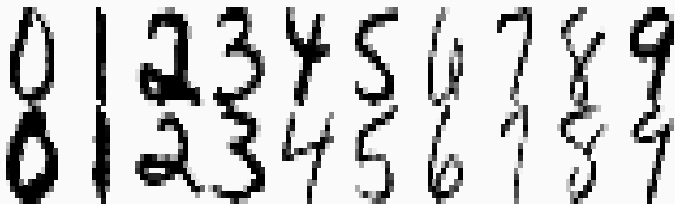
$$d^2(\mathbf{u}, \mathbf{v}) = (\mathbf{u} - \mathbf{v})^T \Sigma^{-1} (\mathbf{u} - \mathbf{v}),$$

where Σ is a covariance matrix.

- Hamming distance is usually used for categorical variables

Example: Handwritten Digit Recognition Data

- Digits 0-9 scanned from envelopes by the U.S. Postal Service
- 7291 training samples, 2007 testing samples
- Apply k NN and calculate the errors
- 1NN with Euclidean distance gives 5.6% error rate
- 1NN with **tangent distance** (Simard et al., 1993) gives 2.6% error



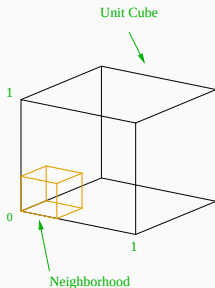
- Need to store the entire training data for future prediction
- Prediction can be **slow**. Needs to calculate the distance from x_0 to all training sample and sort them.
- Some fast nearest neighbor search algorithms such as k-d tree
- A distance measure may affect accuracy

Curse of Dimensionality

- High-dimension low sample size ($p \gg n$)
 - The resolution of the handwritten digit example is $16 \times 16 = 256$
 - Some common imaging data in medical are 1024×1024 while only a few hundred samples are available
 - Strategy games (Go, StarCraft, etc.) may have a huge number of variables
- Curse of Dimensionality
 - For fixed n , as p increases, the data become sparse
 - As p increases, the number of possible models explodes (computation burden, variable selection necessary)

Curse of Dimensionality

- The curse of dimensionality is well illustrated by a subcubical neighborhood for uniform data in a unit cube.
- Suppose the sample points are **evenly spread out on $[0, 1]^p$** , and we want to fit $k = 10$ nearest neighbors with $n = 1000$. Let l be the edge length of the hyper-cube that contains all k -nearest neighbor of a test point. How big is l ?



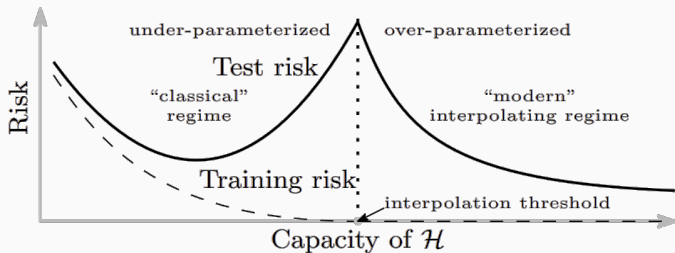
- $l^p \approx \frac{k}{n}$
- When $p = 2$, $l = 0.1$
- When $p = 10$, $l = 0.63$
- When $p = 100$, $l = 0.955$

Low Dimensional Structure

- However, in the previous handwritten digit problem, KNN seems to work pretty well. Why?
- There is potential lower dimensional subspace (manifold).
- Total volume of the data is much reduced — there are more samples within the neighborhood of an existing sample

Double Descent

- Recent research shows that the bias-variance trade-off may not be everything
- E.g., deep learning models are always over-parameterized. However, they still have good performance.



Belkin, et al. "Reconciling modern machine-learning practice and the classical bias-variance trade-off." PNAS (2019)